SREEDARSH T M

Remote sensing | *Satellite Image Processing* | *Research Aspirant in GIS &RS*

Research Portfolio- https://sreedarsh.vercel.app

ORCID +919400668905 | tmsreedarsh@gmail.com | Location (Kanhangad-Kerala-India)

ABOUT ME

Geospatial analyst specializing in remote sensing and GIS for disaster risk, forest monitoring, and agriculture, with work spanning landslide susceptibility, flood mapping, forest fire assessment, coastal vulnerability, and phenology-based crop classification. I build reproducible Python-based pipelines, using AI-assisted development to prototype quickly, and I focus on turning satellite time-series and terrain data into validated, decision-ready results.

EXPERIENCE

July 2026 – Present

Geospatial Analyst | SatSure, Bengaluru, India

- Processing satellite imagery (multispectral, SAR) into actionable insights.
- Building models for crop monitoring, yield forecasting, and climate risk.
- Identifying and classifying features like crop types and land use.
- Creating high-quality reports, maps, and visual outputs for clients.

May 2026 – June 2026

Junior Executive (GIS) | Directorate of Census Operations, Kerala

- Contributing to the 16th Indian Census (Digital Census 2027)
- Collaborating with the MAP Division to maintain data integrity and spatial accuracy for multi-thematic census datasets.

Jan 2025 – June 2025

Project Intern | National Remote Sensing Centre (NRSC), ISRO, Hyderabad, India

- Project: "Burned Area Extraction using AI & ML (Deep Learning).
- Developed an automated, cloud-based data ingestion pipeline in Google Colab using the Earth Engine Python API to efficiently query, filter, and export multi-temporal satellite imagery (Sentinel-2 and Landsat 8/9).
- Engineered interactive GUI widgets to streamline the extraction of 453 specific 31*31 km geographic grids across India, accelerating the preprocessing workflow for downstream machine learning architectures.

ACADEMIC QUALIFICATIONS

DEGREE	UNIVERSITY	SPECIALIZATION / STREAM	YEAR
M. Tech	Bharathidasan University	Geoinformatics	2025
M. Sc	Mangalore University	Geography	2023
B. Sc	Kannur University	Geography Sub: Geology, Statistics	2021
VHSE	Kerala Board of VHSE	ABFS (Science) Agriculture, Biology, Chemistry, Physics	2018

RESEARCH INTERESTS

Areas of Ongoing Learning / Current Learning Goals)

- Remote Sensing, GIS, Digital Image Processing of Satellite Data
- Microwave Remote Sensing (NISAR, SAR, INSAR)
- Python & Earth Engine, Machine Learning, and Deep learning
- Disaster Management/Spatial Analysis

Updated On 16/07/2026

ACCEPTED PAPERS / CHAPTERS

1. SP. Dhanabalan, R. Jegankumar, V. Sindhuja, K. Prakash, **TM. Sreedarsh**. Prioritization of Flood Susceptibility zones using MCDM techniques, TOPSIS, and the CF Model in the Manjalar Watershed: A Part of the Vaigai Basin, South India. DOI PDF

MANUSCRIPTS IN PREPARATION

2. **T.M. Sreedarsh** & Poojith Kumar D P. "Coastal Vulnerability Assessment of Kasaragod and Kannur using Geospatial Techniques." (***Book Chapter)***

ACADEMIC PROJECT

Burned Area Extraction using AI & Deep Learning | *M. Tech Geoinformatics Major Project*

- Applied deep learning algorithms to automate the detection of burned areas from satellite imagery.
- Created a labeled training dataset for identifying burned areas across India to train deep learning models.
- Gained hands-on experience in processing satellite imagery using Google Colab and the Earth Engine API.

Landslide Susceptibility Modeling using Machine Learning | *M. Tech Geoinformatics Mini Project*

- Conducted bivariate analysis and ML techniques to map landslide risks in Kodaikanal and its environs.

Coastal Vulnerability Assessment using Geospatial Techniques | *M. Sc Geography Major Project*

- Analyzed coastal changes and vulnerability indices for the Kasaragod and Kannur regions.

TECHNICAL SKILLS

Programming & Cloud	Python, Google Earth Engine (GEE) API
Development Tools	VS Code, Jupyter Notebook, Cursor, Google Colab, Git & GitHub
GIS Software	ArcGIS and ArcGIS Pro, (DSAS), QGIS, Google Earth Pro
DIP Software	ERDAS IMAGINE
Microsoft Office	Word, PowerPoint, Excel
Others	Generative AI Tools

INVITED TALKS & WORKSHOPS

Resource Person: Geospatial Data Processing with Python | *Oct 2025*

- **Host:** Department of Geography, Kannur University.
- Delivered a one-day hands-on workshop on "**Geospatial Data Processing and Visualization using Earth Engine API & Google Colab**" for the Postgraduate Diploma in Geo-Informatics.

CONFERENCE PRESENTATIONS

- **Site Suitability Analysis for Solid Waste Disposal in Kasaragod Taluk using GIS Techniques**
  - *13th International Conference on Multi-disciplinary Approach for Sustainable Development (UGIT),* Bangalore University | *Dec 2024*
- **Coastal Vulnerability Assessment of Kasaragod and Kannur using Geospatial Techniques**
  - *International Conference on Clean Water, Good Health, Sustainable Cities & Communities (CWGHSCC),* Lovely Professional University, Punjab | *Oct 2023*

WORKSHOPS & ADVANCED TRAINING

- **High-End Workshop (Karyashala) on Hyperspectral Sensing**
  - *Topic:* Agriculture and Water Resources Management.
  - *Organizer:* Accelerate Vigyan Scheme (SERB Initiative).
- **GIAN Workshop on SWAT Modeling (Soil & Water Assessment Tool)**
  - *Topic:* Impact of Landuse/Landcover and Climate Change on Water Resources.
  - *Host:* NIT Trichy (5-Day Workshop).
- **AI for Space Application Training Program**
  - *Host:* Hex-Star Universe | *Feb 2025*
  - Completed 25 hours of intensive training on AI applications in Space Science and Technology.

SEMINAR / CONFERENCE / WORKSHOPS / WEBINAR OVERVIEW

IIRS ISRO E-CLASS	BHUVAN TRAINING	ESRI-MOOC	NASA ARSET	NPTEL	OTHERS
10	3	2	2	-	4

INTERNATIONAL		NATIONAL		WORKSHOP	OTHERS
Journals	Conference	Journals	Conference		Books, Chapters, Monographs, Manuals
-	2	-	-	2	1

Certificates Links …… Click

REFERENCES

I. **M.S.S. Praveen,** ***Scientist/Engineer*** **- 'SE'**, FEG/RSAA, National Remote Sensing Centre **(NRSC)**, Indian Space Research Organization (**ISRO**), Dept. Of Space, Govt. Of India, Hyderabad- -500 037 (T.S.). Mail somasatyapraveen_m@nrsc.gov.in

II. **Dr. R. Jegankumar,** M.Sc., M.Tech., Ph.D., Professor and Head, Department of Geography, School of Earth Sciences, Bharathidasan University, Tiruchirappalli - 620024, Tamil Nadu, Ph: 9894748564, jegankumar@bdu.ac.in

DECLARATION

I declare that the information furnished above is true and correct to the best of my knowledge and belief.

Place: Kanhangad

Signature:

CONTACT

Languages – Malayalam, English, Tamil

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→ +919400668905

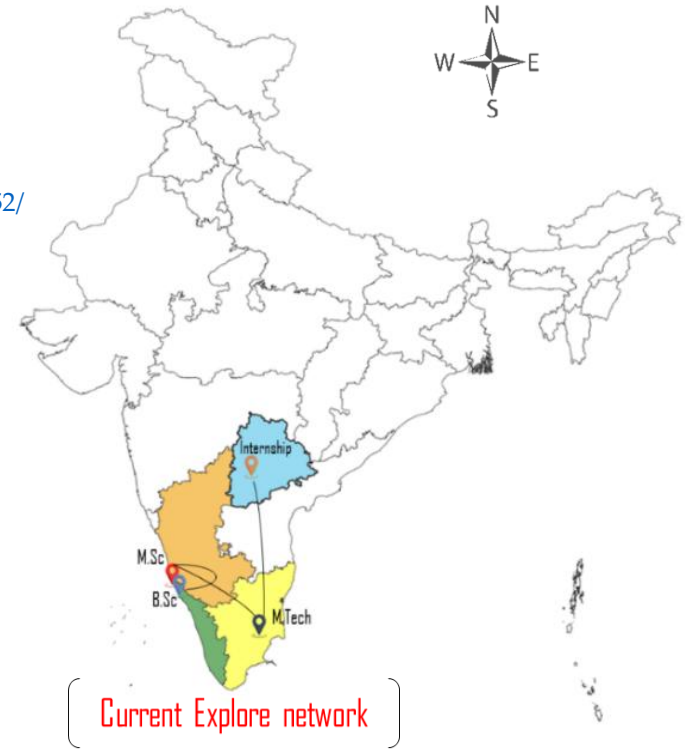
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→ https://scholar.google.com/citations?user=5QaL5pkAAAAJ



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